

A Large-Scale Longitudinal Study of Evolution in URL Length using Common Crawl

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Abstract

The internet and the web are a growing phenomenon. It has quickly gained popularity in 30 years from sharing scientific articles to being available in almost every household for catering to various activities of users. There are many characteristics of the web unknown to the world and very little in-depth studies are available on how the web has changed. We aim to analyse how the length of URL has changed in a span of nineteen years using data from Common Crawl.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Qunoot Khaleeq)

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Chapter 1

Introduction

A Uniform Resource Locator (URL) is a reference used to locate resources on the internet. A URL consists of a scheme, authority, path, and optionally query parameters. Scheme or Protocol is the first part of a URL, with "http" and "https" being the most prevalent instances. Authority aspect often embodies a host name or an IP address, potentially accompanied by a port number. Path refers to a location withing a website's information space segmented by a forward slash ("/"). Query parameters are a set of key-value pairs at the end of a URL, separated from path by a question mark ("?") to help define specific content or actions based on the data being passed [1].

For this project we have assumed that the length of Uniform Resource Locator (URL) has substantially grown since their creation in the early days of World Wide Web. URLs have advanced to incorporate additional information like query strings, filters, encrypted session keys, and other opaque strings assisting users in refining their searches for particular content on a page [1]. In this study, our objective is to examine the primary constituent of a URL and ascertain which element has contributed to its elongation over time. While we have hypothesized that query parameters play this role, our study seeks to validate this assumption.

By scrutinizing a vast array of URLs across various years, we aim to uncover the drivers behind their expansion. The length of a URL holds significance in various aspects of web browsing and online interactions. They can help in the development of intelligent malicious URL detection, compression schemes, caching, page loading speed, search engine optimization, and understanding user click behavior. This study has the potential to open up new avenues of research in the field of web data analysis.

The remaining sections of the paper encompass Background, Implementation, Results and Analysis, and Conclusion.

Chapter 2

Background

According to [2] Google’s index currently covers around 50 billion web pages. However, this is only a rough estimate and does not include the entire web, as not all pages are publicly accessible or indexed by search engines. New pages can be generated in seconds based on personalised user sessions. It is difficult to provide an exact size for the current web, as it is constantly changing and expanding at a rapid pace. This makes analysing the entire web an impossible task.

Common Crawl is an open source petabyte-scale web crawl longitudinal dataset, crawling over 2.5 billion pages per month since 2008 [3]. Each archive is divided into 100 segments. The content of every web page is compressed and stored. Common Crawl dataset is widely used in academic and industrial research. Access to such large web data over a long period of time gives researchers a convenient and cost-effective tool to conduct longitudinal studies which are crucial in tracking changes in content and structure of the web.

Our research aims to thoroughly examine the variations in URL length over time. We will evaluate changes in length of each URL component in relation to last-modified date of crawled pages stored in Common Crawl August 2019 WARC files. The raw data that was collected for previous studies[4] has been stored on a BeeGFS Parallel Cluster File System[5], which is mounted on Cirrus - a High Performance Computing (HPC) facility. This facility has been funded by the University of Edinburgh and the Engineering and Physical Sciences Research Council (EPSRC). The data provides us with the necessary resources for our research.

Similar to the majority of High-Performance Computing (HPC) services, Cirrus employs Slurm, a job scheduler to manage its resources. Slurm is an open-source sophisticated job scheduling and workload management system used in high-performance

computing environments. It efficiently allocates and manages computing resources, such as processors and memory, to ensure optimal utilization and performance. Slurm enables users to submit, monitor, and control jobs, making it easier to distribute tasks across a cluster of computers. Its flexibility accommodates a wide range of job types, from simple to complex parallel computations [6].

2.1 Data

Starting from 2013, Common Crawl employs the Web ARChive (WARC) format to store data from all crawls. Each WARC file is around 1GB (compressed) in size. Within each segment of August 2019 crawl, there are 560 WARC files, and each month comprises 100 segments, numbered 0 to 99. Each WARC file contains information for over 50,000 web pages. These pages encompass data on successful requests (WARC-Type: request), raw HTTP responses from websites (WARC-Type: response) inclusive of headers and original content mostly of type HTML, and metadata concerning the crawl process (WARC-Type: metadata). Below is a representative response excerpted from a WARC file.

```
WARC/1.0
WARC-Type: response
WARC-Date: 2019-08-17T22:50:00Z
WARC-Record-ID: <urn:uuid:9437ec6d-b5c9-4bf5-8ebd-d1e4afaba126>
Content-Length: 76444
Content-Type: application/http; msgtype=response
WARC-Warcinfo-ID: <urn:uuid:26f6c2e8-fa74-4ec7-8eb2-408431694074>
WARC-Concurrent-To: <urn:uuid:a72780fa-6841-43a2-99f3-89ef2348cb10>
WARC-IP-Address: 199.233.239.106
WARC-Target-URI: http://00441.net/209973/
WARC-Payload-Digest: sha1:ALF243SFDRCZMC2C6VOH6JW52534PD4
WARC-Block-Digest: sha1:427INNMBFVXJ2UIGKONW2RVR5OYMUCX6
WARC-Identified-Payload-Type: text/html

HTTP/1.1 200 OK
Content-Type: text/html
X-Crawler-Content-Encoding: gzip
Last-Modified: Tue, 16 Jul 2019 07:59:15 GMT
```

```
ETag: "90a4c-57280b173693a"  
Accept-Ranges: bytes  
Server: Microsoft-IIS/7.5  
X-Powered-By: ASP.NET  
Content-Length: 76106
```

```
<!DOCTYPE html>  
<html>  
<head>  
<meta charset="utf-8">  
...  
<a href="/map.html" target='blank' >Article</a>  
</body>  
</html>
```

In addition to WARC files, Common Crawl archives WAT, WET, and index files which are much smaller in size. WAT files contain computed metadata for WARC-stored data, while WET files store extracted plaintext from the WARC data. Common Crawl Index is a searchable catalog of URLs and metadata from web crawls, enabling targeted data retrieval.

Given the time, and resource constraints, working with archives from different periods is unfeasible. However, to maintain the longitudinal aspect of this study, we make use of last-modified dates and Target URI. The WARC-Target-URI consistently resides in the WARC header of a response record. However, it's important to note that the presence of Last-Modified in the HTTP header is not always guaranteed; its occurrence varies. Given that WARC files exclusively include response headers, we will utilize them for our analysis. August 2019 archives were arbitrarily chosen. This collection encompasses 56,000 WARC files, with a compressed size of 54TB, ensuring a comprehensive dataset for analysis.

2.2 Resources

This work used the Cirrus UK National Tier-2 HPC Service at EPCC funded by the University of Edinburgh and EPSRC (EP/P020267/1). Cirrus compute provision consists of 280 standard nodes. Every standard compute node is comprised of two 18-core

series processors. Each of these nodes possesses a combined memory of 256 GB, which is shared between the two processors [7]. For this research, we have been assigned 100,000 CPU hours on Cirrus.

Running extensive tasks on login nodes is prohibited. To utilize additional nodes, we employed sbatch to submit batch scripts to slurm. The batch script incorporate "#SBATCH" options preceding executable commands. Once a non-comment, non-whitespace line is encountered, sbatch halts processing further #SBATCH directives. Upon successful transfer to the Slurm controller and allocation of a job ID, sbatch exits. However, the script doesn't instantaneously receive resources; it could sit in the pending job queue until the needed resources are accessible [8]. To process data files we used approximately 14,256 CPU hours by running various tasks in parallel. To ensure that a task remains confined to a single node, we allocated 18 CPU cores per task. Memory allocation is determined by multiplying number of requested CPU cores by 1/36th of the total available memory, allowing for a maximum of 64 GB of RAM for every task, proving effective for handling multiple WARC files simultaneously.

The WARC files required for this study are stored using BeeGFS. It is a parallel file system designed and developed specifically for high-performance computing environments. it includes a distributed metadata architecture for scalability and flexibility reasons. Its most used and widely known attribute is data throughput [9]. Furthermore, while there is no explicit limitation on disk usage for the file system, we have allocated 256 GB of disk storage for this project, which has been utilized to store the processed data.

2.3 Previous Works

There has been very little work done on analysing properties of the internet in recent years. In 2005 [10] analysed URL length as one of the characteristics of the Web of Spain. The authors report that there were very few lengthy URLs found in the data they had collected using a crawler. The result show that the average length of a URL including protocols and parameters was 67 characters. It regarded URLs longer than 150 characters as anomalies, that did not provide any content inaccessible using shorter URLs couldn't. [11], [12] Studies prior to that show an average URL length of 50 to 62 characters. Most websites had a depth of 2.

[13] research focused on the correlation between URL length and their persistence. They found that URLs under 50 characters tend to be more enduring compared to

longer ones. Dynamic URLs, which averaged 77.1 characters, exhibited less persistence than static URLs, which averaged 49.2 characters. Their observations indicated that excessively long URLs are prevalent in poorly designed websites that undergo rapid restructuring or deactivation. Typically, URLs situated at shallower levels tend to attract more visits. Their observations showed that the depth of a URL didn't impact its persistence. Through their analysis of persistent URLs, they noted that these URLs could be located at varying depth levels depending on the website's structure.

Authors of [14] found that malicious advertisement URLs exhibit a wider distribution of lengths compared to benign URLs. Most benign URLs fall within the 25 to 50 character range, with an average of 44.28 characters. A benign URL typically includes an average of 8.64 special characters and a path length of approximately 17.54 characters. In contrast, the majority of malicious ad URLs range from 25 to 105 characters, with an average length of 63.14 characters. These malicious URLs contain an average of 13.98 special characters and a path length of about 42.60 characters. Additionally, less than 2 percent of malicious ad URLs employ IP addresses as domain names.

These findings collectively underline the significance of URL length in shaping online experiences, security considerations, and web design strategies.

Chapter 3

Implementation

The raw data is located within the beegfs directory, while the code resides in a separate filesystem accessible from both login and compute nodes. Since administrative privileges weren't available on Cirrus for package installation, all code was executed in a Singularity [15] Debian container, simplifying the installation and utilization of necessary packages. For parallel data processing, a bash sbatch script was authored to submit tasks to the scheduler. This script incorporates essential #SBATCH directives and a single srun command for launching parallel tasks. There are 3 mandatory sbatch properties that require specification for every job: partition referring to type of node to run on, Quality of Service(QoS) which specifies the job limits that apply, and account code which will be charged from its budget. The script is designed in a generic manner, using standard Quality of Service (QoS) and partition settings. It is versatile and employed at various stages to initiate distinct jobs. The standard Quality of Service places no limit on the number of jobs per user, but enforces a maximum queue wait limit of 500 jobs, with up to 88 nodes (equivalent to 3168 cores). The standard partition simply corresponds to CPU nodes. Additional #SBATCH directives are tailored to each job and are defined in the command line for job submission. Typically, we specified parameters such as the number of tasks, number of CPUs per task, desired output file name, and maximum walltime required for the job. In case these options are not specified, default values will be applied for each parameter. This includes 1 task, 1 CPU core, "slurm_<JobID>.out" as the output file name, and a maximum walltime of twenty minutes. To run the script from command line, state the "sbatch" keyword followed by job-specific sbatch arguments, the filename, and the input arguments the file expects. Upon submission, a JobID is provided. To view details about job status, node allocation, and node names, use the squeue command. Jobs typically progress through

the following states: PENDING, RUNNING, COMPLETING, and COMPLETED.

3.1 Data Pre-processing

A total of 56,000 WARC files spread across 100 segments from week 35 of 2019 were utilized for extracting pertinent data. These common crawl files adhere to a specific naming structure: CC-MAIN-YYYY-WW/[segment] /path/CC-MAIN-[file-number].warc.gz. Here, "WW" signifies the week within the year when the data for that particular month is stored, and "file-number" ranges from 0 to 559. A Slurm job has the capability to simultaneously execute multiple tasks, identified by IDs ranging from 0 to ntasks-1. In our setup, each job was configured to run 16 tasks, with each task utilizing 18 CPU cores and occupying half a node. This collectively resulted in 288 CPU cores being distributed across 8 full nodes. Each task undertook the processing of 35 files, all of which concluded with the same file-number, originating from various segments. To mitigate thrashing, each task is structured to complete a maximum of 8 files at a time.

The actual preprocessing procedure starts with unzipping a file, followed by piping the output to remove carriage returns. This step is crucial for enabling the proper functioning of the data extraction expressions. Subsequently, the output is piped once more to facilitate data extraction through 'awk' pattern matching, focusing on lines initiated with "WARC-Type: response" and continuing until the occurrence of a second blank line. This process selects two distinct data blocks containing "Target-URI" and "Last-modified date". The output is then subjected to further pattern matching using 'grep,' which isolates only the "WARC-Target-URI" and "Last-Modified" lines. The output is piped again to establish pairs of "Target-URI" and "Last-modified date," separated by a tab space. Following this, another 'sed' pattern matching step is employed to eliminate a secondary occurrence of the "Last-modified" date from the same page. This precautionary measure is necessary due to the absence of standardized headers and responses documentation within the Common Crawl dataset. Finally, the output is directed to a temporary directory, taking advantage of its location in RAM to expedite read-write access. Subsequently, it is gzipped and saved in a personal working directory, and then cleared from the /tmp.

Jobs were submitted in various batches to the scheduler. The initial two jobs handled one segment each, with each task processing 35 files, and both tasks concluded in 24 minutes and 38 seconds. In the third job, 5 segments were pre-processed completing in 2 hours, 2 minutes, and 49 seconds. The fourth job processed 6 segments in 2 hours,

31 minutes, and 3 seconds, while the fifth managed 7 segments in 2 hours, 51 minutes, and 1 second. The sixth job dealt with 10 segments over 4 hours, 3 minutes, and 47 seconds. Moving on, the seventh job processed 15 segments in 5 hours, 56 minutes, and 3 seconds. The eighth job addressed 19 segments in 8 hours, 1 minute, and 23 seconds. Lastly, the ninth job pre-processed 33 segments in 13 hours, 56 minutes, and 58 seconds. All segments completed the first stage of pre-processing, although some segments were inadvertently deleted later in the project. Unfortunately, due to a lack of backup, a separate job was run for a list of 21 segments.

3.2 Data Preparation

The initial stage of pre-processing consumed the most time and required significantly more resources compared to the subsequent stages. We now possessed files that are somewhat more organized and much smaller in size, enhancing their usability. Each row contained a Target-URI; however, not every row includes a Last-modified date due to its absence in the response headers of the original data. These refined files were stored in a directory structure akin to that of the raw data.

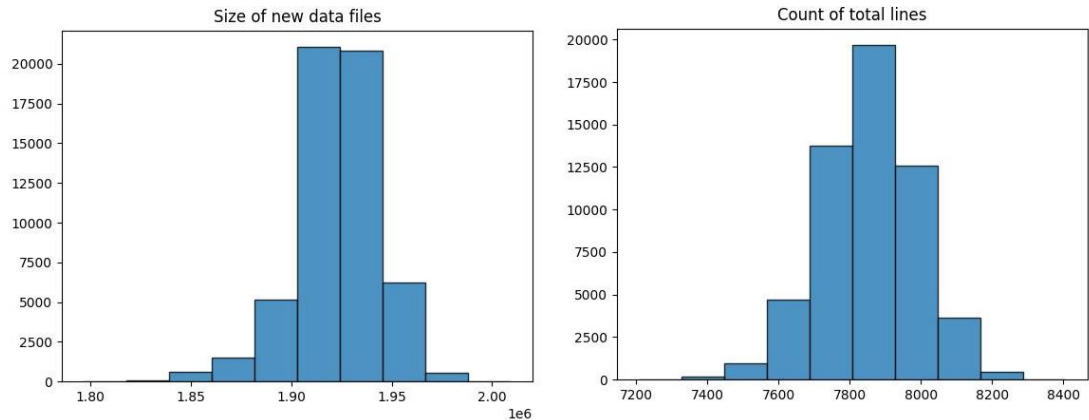


Figure 3.1: Compressed size (in bytes) and count distribution of new data

As depicted in Figure 3.1, it is evident that the smallest file measures 1.78 MB, while the largest reaches 2 MB. On average, files are approximately 1.92 MB in size, with a variation of 0.2 MB. Around 50 percent of the files fall within a range of 1.91 MB to 1.94 MB. Regarding the line count, a file has at least 7206 records of Target-URI, and at most 8408. However, not every line necessarily includes a Target-URI and Last-Modified date pair. On average, there are around 7855 lines per file, with a fluctuation of 135 lines. Less than 5% of the files have fewer than 7720 URI records,

and fewer than 5% exceed 8072 lines. This indicates a relatively compact distribution and consistent file characteristics.

Given the longitudinal nature of our work, we refined our dataset by excluding lines that only featured the Target-URI without its corresponding Last-Modified date. Due to inconsistent date formats in the Last-Modified field, it was unsuitable for analysis in its original form. To address this, we employed the "email.utils" module in Python to recognize diverse date formats and subsequently convert these dates to epoch (seconds since 1 January 1970) format. Pairs featuring date formats unrecognized by the library were excluded due to their limited occurrence and the inefficiency of employing workarounds to handle them. Additionally, we utilized the "urlsplit" function from the "urllib.parse" library to dissect the URL into its constituent parts: authority, path, and query. Consequently, our revised TSV files contained fields for URL, URL length, authority length, path length, query length, and time in epoch format.

Subsequently, we aggregated each segment into a single file and sorted the data based on the last-modified date. As a result, we obtained 100 TSV files, encompassing the information of each segment. A few of the last-modified dates were negative, some even dated back to 1970, and a handful indicated dates beyond August 2019 – anomalies that were of minimal occurrence. To refine the dataset, we filtered out data points outside the range between 2000 and August 2019. Additionally, we combined sets of 10 segments together. Unfortunately, due to time constraints and the unavailability of specialized databases or frameworks for handling extensive data, we had to rely on sorting data chronologically and merging it. Despite this, the total merged dataset amounted to 55GB, making it infeasible to load entirely into memory for creating summaries and graphs. To mitigate this challenge, we partitioned the data into smaller units based on year-month groups, rendering it more conducive for our analytical endeavors.

3.3 Visualisation Technique

Ultimately, for the purpose of visualizing and effectively summarizing the gathered data, we harnessed the capabilities of the pandas, seaborn and matplotlib libraries in the Python programming language. By iteratively reading each month's file, we not only extracted the necessary information but also conducted comprehensive data summarizing. This allowed us to construct insightful graphs that offer an overarching view of the data distribution, enabling us to identify potential patterns and trends that might be present within the dataset. The graphical representation proved instrumental

in enhancing our understanding of the evolving characteristics of URL length and its constituent elements over time.

3.4 Challenges

We faced certain challenges during the implementation of our large scale longitudinal study.

- **Data Accuracy:** To ensure accuracy and completeness of processed data, we implemented a continuous verification process. This involved comparing various aspects of data such as, file sizes, total lines, URIs, and last modified dates to ensure it meets expected information. Additionally, we carefully examine the output log files for any indications of bugs, errors, or unexpected behavior. We compared the early processed files with previous works to validate the correctness of our program. Through iterative updates to the code, we made necessary adjustments until we achieved satisfactory results.
- **Time and resource constraints:** Given the limited compute time and storage available on Cirrus, it was crucial for us to optimize resource utilization while ensuring timely completion of our work. To achieve this balance, we extensively tested multiple parallelization methods within SLURM. Through careful adjustments of the number of jobs, tasks, CPUs, and nodes, we aimed to find the optimal configuration that maximized efficiency and allowed us to accomplish our tasks within a reasonable time frame. Due to resource sharing, there were occasions when the resources we required were temporarily unavailable. This could be due to maintenance activities, higher priority jobs taking precedence, or reserved resources that were not accessible to us. In such situations, we had no choice but to patiently wait until the resources became available again. These unforeseen delays were beyond our control.
- **Technical challenges:** Due to limited control over the resources, we encountered challenges in gaining a comprehensive understanding of how to maximize their potential. This lack of control made it difficult for us to fully leverage the available resources to their fullest extent. As a result, we had to work within the constraints and limitations imposed by the system.

Chapter 4

Result and Analysis

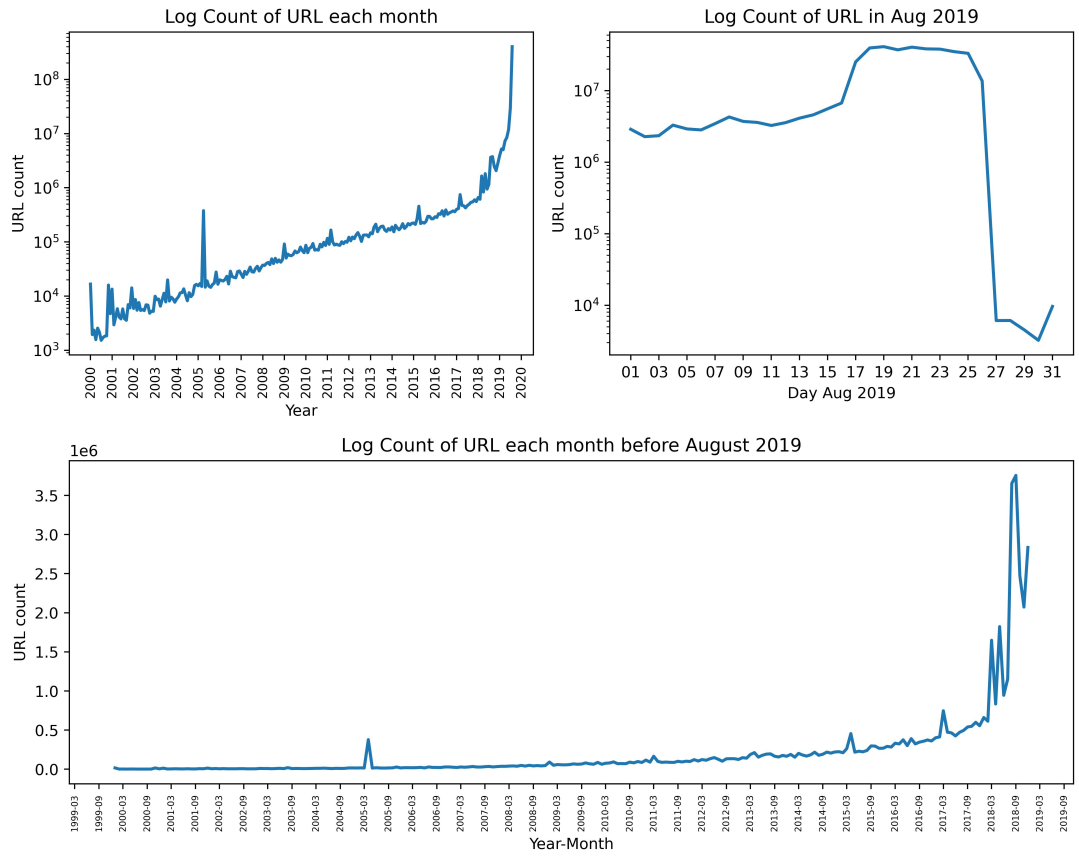


Figure 4.1: Number of URLs over Time

Currently, our dataset comprises over 520 million instances of Target-URI and Last-Modified date, spanning from January 2000 to August 2019. Upon closer examination of the data distribution, a noticeable trend emerges – a majority of the URIs are associated with a last-modified date in August 2019. This phenomenon can be attributed to the

contemporary web landscape where pages are frequently generated dynamically within seconds. This pattern aligns with the notion that older years contain fewer Target-URIs, likely due to the transient nature of many web pages on the internet. In fact, the dataset's size for the month August 2019 alone exceeds 402 million lines, surpassing the cumulative data from 2000 to July 2019, which amounts to approximately 118 million lines. This discrepancy raises concerns about data bias and highlights the need for careful consideration when analyzing and interpreting the dataset.



Figure 4.2: Average Length across Months

The combined overall mean URL length is 79.634 characters, while the average authority length is 18.348 characters, path length averages at 39.049 characters, and query length has an average of 14.510 characters. Specifically for August 2019, the mean URL length stands at 82.635 characters, with an average authority length of

18.383 characters, path length averaging 40.336 characters, and query length averaging 16.112 characters. Examining Figure 4.2, we observe intriguing anomaly in August of 2013, exhibiting a significant spike in the average query and URL length. Similarly, there is an unusual surge in path length in September of 2009. We will delve deeper into the analysis of these two cases for further investigation. On the whole, there has been a consistent rise in URL length. Table 4.1 presents information about the average change and percentage change in the number of characters, relative to the overall mean.

Table 4.1: Average and Percentage change in URL length

Element	Average change	Percentage change
URL	11.199	14.06 %
Authority	1.001	5.45 %
Path	4.932	12.63 %
Query	10.342	71.27 %

The length of the authority segment has remained relatively stable, showing only a slight increase of 1 character each month. However, the path length has shown a more noticeable growth, with an average change of 4.932 characters per month. Despite the similarity in average changes between overall URL and query length, the percentage of change in the query length, relative to the overall mean, is unexpectedly high. We will look into this aspect further in the subsequent sections of our study.

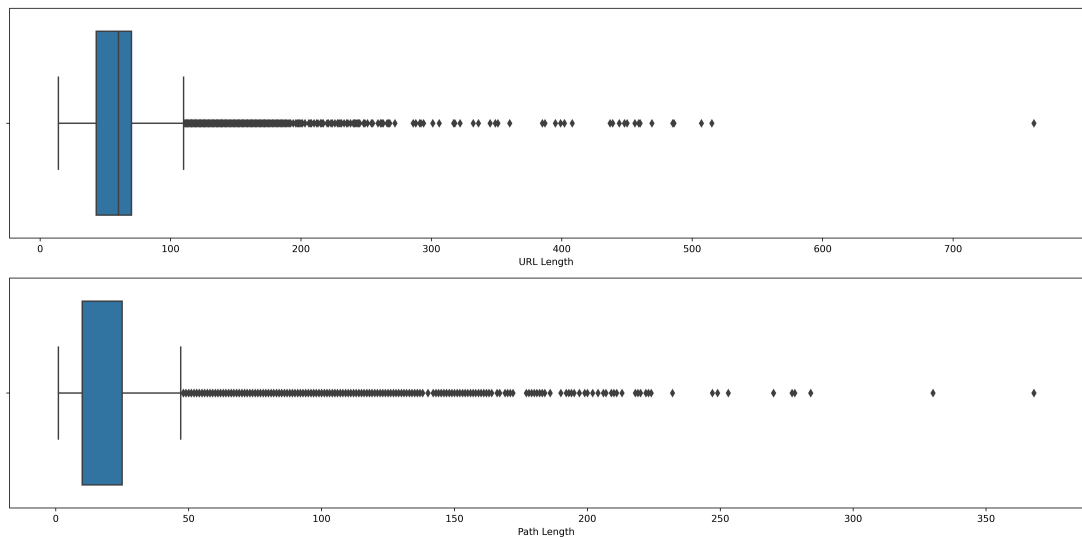


Figure 4.3: Boxplot URL Length and Path Length in September 2019

Upon a thorough examination of the values from September 2009, where Figure 4.2 shows a significant spike for both URL length and path length, we noticed a resemblance in their distributions. Figure 4.3 depicts the distribution of path and URL lengths in September 2009. Outliers are marked with black diamonds. The presence of extreme outliers in the path length has pushed the median towards one end of the distribution, causing it to be less representative of the data's central tendency. Furthermore, there isn't considerable variability within the middle 50 percent of the data. Around 25 percent of the dataset, comprising approximately 155,561 values, exhibit values significantly exceeding the upper quartile value of 55. This exerts a substantial impact on the mean. In contrast, the URL length displays a distinct median, signifying a more evenly distributed dataset compared to the path length.

From this observation, we deduced that the presence of outliers in the path length data was responsible for the spike observed in URL length. The outliers extended from 56 to 828 characters. This deduction was supported by the fact that the path length demonstrated an unusual standard deviation of 111.63 characters, contrasting with the standard deviations of 5.13 and 6.08 characters for authority length and path length, respectively. To further assist our inference, below is an illustration of a URI featuring an extended path comprising 416 characters. The apparent repetition of characters might arise from the constraint that URLs are transmitted through the Internet solely using the ASCII character-set. URL syntax also requires some punctuation characters to be escaped, e.g. `'i'` (hex 3C), `'&'` (3E) and space (20). Since URLs frequently encompass characters beyond the boundaries of the ASCII set, they need to be transformed into a valid ASCII format. URL encoding accomplishes this by substituting unsafe ASCII characters with `"%"` followed by two hexadecimal digits[16].

```
http://www10.giscale.com/nbc/articles/1/737477/%3Ci%3EAutodesk%
20Flame%3C/%3Ci%3ESid%20the%20Science%20Kid%3C/%3Ci%3EHouse%3C/%3C
i%3E3D%20%20design%3C/%3Ci%3EG.I.%20Joe:%20The%20Rise%20of%20the%2
0Cobra%3C/%3Ci%3ESmoke%3C/%3Ci%3EAutodesk%20Inferno%3C/%3Ci%3ESupe
r%20Bowl%202009%20ads%3C/%3Ci%3EFlare%3C/%3Ci%3EFlint%3C/&lt;i&gt;
House&lt;/&lt;i&gt;http://www.fxphd.com/courses&lt;/&lt;i&gt;G.I.%2
0Joe:%20The%20Rise%20of%20the%20Cobra&lt;/i&gt;
```

After decoding the URL, it became evident that the URL was originally short. However, the extended path contained spaces and spacial characters. Similar characteristics were observed in other URLs with lengthy paths, which also contained script other than English. Thus, it was reasonable to conclude that the unusually long URLs from

September 2009 were indeed anomalies.

<http://www10.giscale.com/nbc/articles/1/737477/>*Autodesk Flame*
Sid the Science Kid
House
3D design
G.I. Joe: The Rise of the Cobra
Smoke
Autodesk Inferno
Super Bowl 2009 ads
Flare
Flint
House
<http://www.fxphd.com/courses>
G.I. Joe: The Rise of the Cobra

Continuing our investigation, we delved into the data from August 2013, where Figure 4.2 shows a spike in URL and query length. Figure 4.4 presents unexpected outcomes concerning the distribution of data in query length. Notably, the lower quartile and median are equivalent, and the upper quartile is in close proximity. This pattern suggests a high concentration of data around a single value. Approximately 75 percent of the data had a path length of zero, meaning they lacked a path component in their URLs. As a consequence, any URI with a path component became an outlier. The distribution of data for URL length is positively skewed, and it also shows a large presence of outliers. The outlier pattern observed closely resembles the pattern seen in the outliers of query lengths. Strikingly, we found that this pattern persisted throughout all the months. Interestingly, despite this consistent behavior, the query length from August 2013 remained distinctive compared to other periods. To comprehend the reason behind the significantly larger query strings for August 2013, we once again examined a few examples.

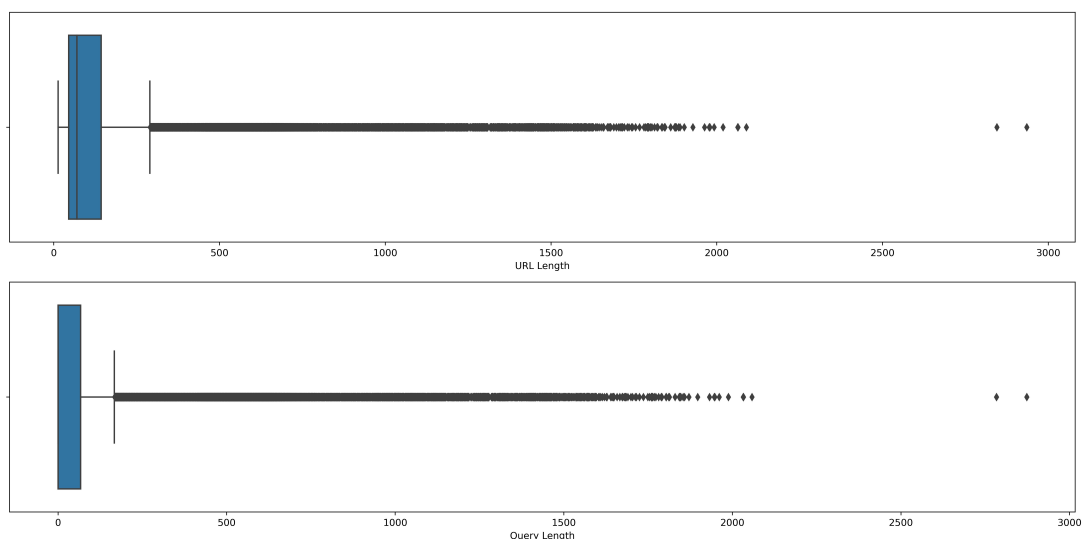


Figure 4.4: Boxplot URL Length and Query Length in August 2013

This scenario turned out to be analogous to the previous example, albeit in this case, it pertained to the query string. Although query strings can be lengthy, what made

August 2013 distinct from other months was the inclusion of non-ASCII characters. We have come to understand that unusually long elements of the URI are primarily attributed to the presence of spaces, special characters, and non-English letters.

```
http://www.posterous.com/share?linkto=https%3A%2F%2Fwww.theywork
kforyou.com%2Flords%2F%3Fid%3D2019-07-11c.1886.9%26amp%3Bp%3D10088%
23g1890.1&title=UK%26%238217%3Bs%20Ambassador%20to%20the%20USA%3A%
20Resignation%20-%20%26lt%3Bi%26gt%3BStatement%26lt%3B%2Fi%26gt%3B
%20%7C%20Lords%20debates&selection=My%20Lords%2C%20is%20not%20Sir%
20Kim%20the%20only%20person%20to%20end%20with%20any%20credit%20in%
20this%20matter%3F%20The%20person%20who%20leaked%20the%20details%2
C%20the%20President%20and%20the%20overweening%20ambition%20of%20Mr
%20Boris%20Johnson%20are%20very%20happy
```

Here is the original string once it's decoded.

```
http://www.posterous.com/share?linkto=https://www.theyworkfor
ou.com/lords/?id=2019-07-11c.1886.9&p=10088#g1890.1&title=UKs Amba
ssador to the USA: Resignation - <i>Statement</i> | Lords debates&s
election=My Lords, is not Sir Kim the only person to end with any c
redit in this matter? The person who leaked the details, the Pres
ident and the overweening ambition of Mr Boris Johnson are very hap
py
```

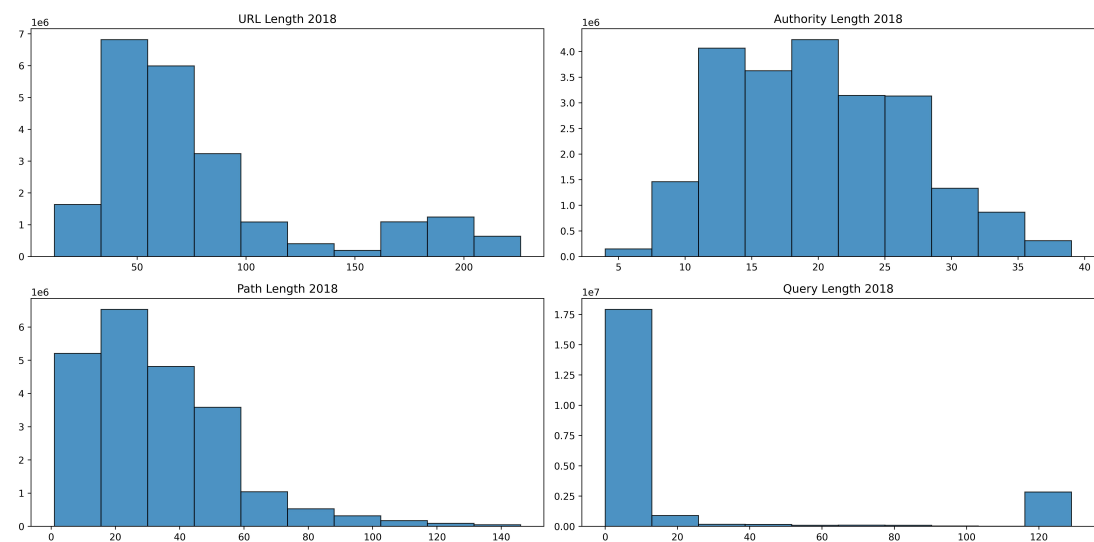


Figure 4.5: Distribution of data points from year 2018 excluding outliers

Our next step involves a thorough and detailed analysis of data distribution in 2018. This choice is attributed to several reasons: it encompasses an entire year's worth of data,

being a recent year, it offers a larger dataset (22 million data point) compared to earlier years, and it avoids the potential complications arising from the significant exponential increase in sample size that is observed in the year 2019. The histogram illustrated in Figure 4.5 captures 99.5 percent of the data, while excluding outliers with notably large values. These outliers can exert a substantial influence on the mean, pulling it away from the median towards the tail of the distribution. The calculated mean URL length is 80.55 characters, contrasting with the median length of 64 characters. Upon removing these outliers, the distribution of overall URL length displays an asymmetrical pattern. A substantial portion of URL lengths, approximately 6.8 million or about 40 percent of the dataset, falls within the range of 33.4 to 54.8 characters. Furthermore, the middle 50 percent of URL lengths span from 46 to 89 characters.

Shifting our focus to authority length, the distribution appears to be relatively balanced, albeit slightly skewed towards the right. The bulk of the values cluster within the ranges of 18 to 21.5 characters and 11 to 14.5 characters, with approximately 4.2 million and 4 million data points, respectively. The mean authority length stands at 19.925 characters, closely mirroring the median value of 19 characters.

In terms of path length, we observe it is positively skewed, with a substantial majority of values (approximately 6.5 million, constituting around 30 percent of the dataset) falling within the range of 15.5 to 30 characters. Notably, as we previously uncovered, 75 percent of URIs exhibit no query component. Among the remaining 20 percent of the data, the majority of query lengths are concentrated within two ranges: 116.1 to 129 characters and 12.9 to 25.8 characters. These two ranges account for approximately 12.6 percent and 4.11 percent of the total dataset, respectively. Exploring the remaining 0.5 percent of the data will offer us a deeper understanding of query length and its distribution characteristics.

Table 4.2 reveals a striking similarity in the number of outliers between URL and Query lengths across each interval. Notably, by incorporating the mean authority length and path length into each query interval, we observe a resemblance between these intervals. This observation resonates with our earlier findings, where we discovered that lengthy URLs had repetitive encoded characters. However, Upon scrutinizing instances with substantial length, we noticed that not all repeated characters were solely attributable to the presence of non-ASCII characters. Some instances contained timestamps and encryption strings as well. Here is an instance of a URI with a length of 1501 characters. Among these, 1395 characters belong to the query component, while the authority and path lengths are 31 and 66 characters, respectively.

Table 4.2: Outliers from 2018

URL length	URL Count	Query Length	Query Count
228.	109,396	131.	85638
718.2	1166	624.6	1109
1208.4	178	1118.2	177
1698.6	101	1611.8	100
2188.8	19	2105.4	21
...	...	...	...
5130.	5	5067.	5

[https://chronus-mentor.s3.amazonaws.com/programs/ckattachements/27412/original/Champs%20Plans%20Flyer.pdf?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=ASIAVRAXBKKODB47SXUF%2F20190817%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20190817T223647Z&X-Amz-Expires=900&X-Amz-Security-Token=AgoJb3JpZ2luX2VjEM7%2F%2F%2F%2F%2F%2F%2F%2F%2FwEaCXVzLWVhc3QtMSJHMEUCIBjGeGFUENIr%2BcRnBi499s7Zu5N4FgpG695FguhpAhvCAiEAiOdbkYYtXcUC2YSrekHubX6NU5s5eGyHr6KyP%2Bm%2Fqjsq2gMIZhAAGgwzODAxNTI5MjY4NzYiDKDSB5y2kIQtdKDMxyq3AwXNA4ip6RzFvjXU4Uyp%2BTOL2rSq0VOZXq4%2F17m5BRyBKC4LJxV%2FmAo3L4tOLbqJaUn%2FxicE7anonbXww4xOtLZIVcLMuFd7gap83nxx0Jq5cg84tthbuhp4yHXlo7x10GV35PsOIX4vDNE8866Uh3jHkj%2FSH9N0bliEwTMwmNjNB3w9GkSkqnocPDVC%2BM2FNJT7BztNhZofGK%2Ba7pbWH7BAtaX4ky8GS%2BgngILLm1DqDatxvp4OcA7PcxSka4g7mmibVXYcIb107ucOEQMIYPC0EJ%2Bd%2Ffu4qyFHrgINwrznFbHnLaN%2BUBP6xk%2FFbtUjgKAXmVVBkwE8ZMVf4z7exWZ8NpXRW1YJa6AixJ4Z2jpPolmO87oLW8aA0Wunbi%2BLhdpNdS1NNiFvADELiTn7T83eSg8hQulU9rrHpTglW1NqYmW%2BxS5gkE6MZASc8r%2BySAUAa9sm2vTR6DvrZWqAYrSa0bFbnC1lJs529QYvGYpdHgG6oPO0vMmBG0c30ulT74YgQD%2Bd9aPR3fyuDs4N58%2B6iY%2F10jETSDtXvV87d1nKjWC5SiF7omCJOsw2PvnIZdrTHyAJ8widzh6gU6tAFSICchg2iYNp771KjS7p%2Fb4ZaT34RU8WITShygfaAWYoKx1JUUVGX17eDhO4QoN1GvSfMuFkFQFif3y1m8CxXi0qEo88FRUMRYgeOlPOQwPZstjDMzxXwVhOMdssn%2F3V%2FH0QGdQD5a6V43Ebq5W0zX737Whi91rX%2BnxjhiuFksdy9W4Lp3Yta%2Fa7uou1ddBLEHnfBLmARwoC%2BonJy8NEGyIhUFWJduZYQfaezjdWENmT0vB1eA%3D&X-Amz-SignedHeaders=host&X-Amz-Signature=704bb6dd3dfa297bc0c6d94d19e3ac61b8aaba2bb99a77ca77f57a805b357ef2](https://chronus-mentor.s3.amazonaws.com/programs/ckattachements/27412/original/Champs%20Plans%20Flyer.pdf?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=ASIAVRAXBKKODB47SXUF%2F20190817%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20190817T223647Z&X-Amz-Expires=900&X-Amz-Security-Token=AgoJb3JpZ2luX2VjEM7%2F%2F%2F%2F%2F%2F%2F%2F%2F%2FwEaCXVzLWVhc3QtMSJHMEUCIBjGeGFUENIr%2BcRnBi499s7Zu5N4FgpG695FguhpAhvCAiEAiOdbkYYtXcUC2YSrekHubX6NU5s5eGyHr6KyP%2Bm%2Fqjsq2gMIZhAAGgwzODAxNTI5MjY4NzYiDKDSB5y2kIQtdKDMxyq3AwXNA4ip6RzFvjXU4Uyp%2BTOL2rSq0VOZXq4%2F17m5BRyBKC4LJxV%2FmAo3L4tOLbqJaUn%2FxicE7anonbXww4xOtLZIVcLMuFd7gap83nxx0Jq5cg84tthbuhp4yHXlo7x10GV35PsOIX4vDNE8866Uh3jHkj%2FSH9N0bliEwTMwmNjNB3w9GkSkqnocPDVC%2BM2FNJT7BztNhZofGK%2Ba7pbWH7BAtaX4ky8GS%2BgngILLm1DqDatxvp4OcA7PcxSka4g7mmibVXYcIb107ucOEQMIYPC0EJ%2Bd%2Ffu4qyFHrgINwrznFbHnLaN%2BUBP6xk%2FFbtUjgKAXmVVBkwE8ZMVf4z7exWZ8NpXRW1YJa6AixJ4Z2jpPolmO87oLW8aA0Wunbi%2BLhdpNdS1NNiFvADELiTn7T83eSg8hQulU9rrHpTglW1NqYmW%2BxS5gkE6MZASc8r%2BySAUAa9sm2vTR6DvrZWqAYrSa0bFbnC1lJs529QYvGYpdHgG6oPO0vMmBG0c30ulT74YgQD%2Bd9aPR3fyuDs4N58%2B6iY%2F10jETSDtXvV87d1nKjWC5SiF7omCJOsw2PvnIZdrTHyAJ8widzh6gU6tAFSICchg2iYNp771KjS7p%2Fb4ZaT34RU8WITShygfaAWYoKx1JUUVGX17eDhO4QoN1GvSfMuFkFQFif3y1m8CxXi0qEo88FRUMRYgeOlPOQwPZstjDMzxXwVhOMdssn%2F3V%2FH0QGdQD5a6V43Ebq5W0zX737Whi91rX%2BnxjhiuFksdy9W4Lp3Yta%2Fa7uou1ddBLEHnfBLmARwoC%2BonJy8NEGyIhUFWJduZYQfaezjdWENmT0vB1eA%3D&X-Amz-SignedHeaders=host&X-Amz-Signature=704bb6dd3dfa297bc0c6d94d19e3ac61b8aaba2bb99a77ca77f57a805b357ef2)

Figure 4.6 illustrates the average query length after removing URLs with no query

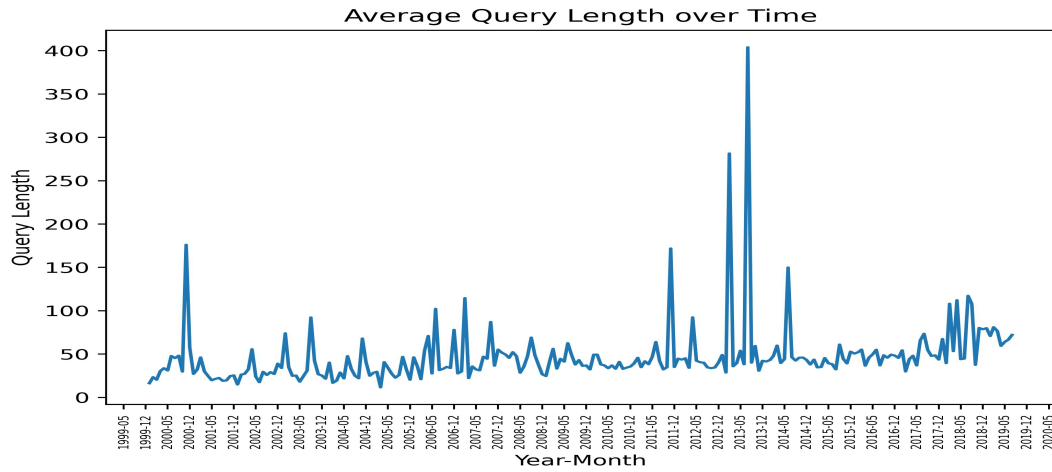


Figure 4.6: Average number of characters in URL query over time

component. While the pattern remains comparable to the previous graph depicting query averages, the actual average values have changed. We have already investigated cases of anomalies that contribute to the spikes observed in specific years. We are now interested in exploring the relationship between URL length and each of its components: authority, path, and query, with the use of scatter plot with linear regression fit.

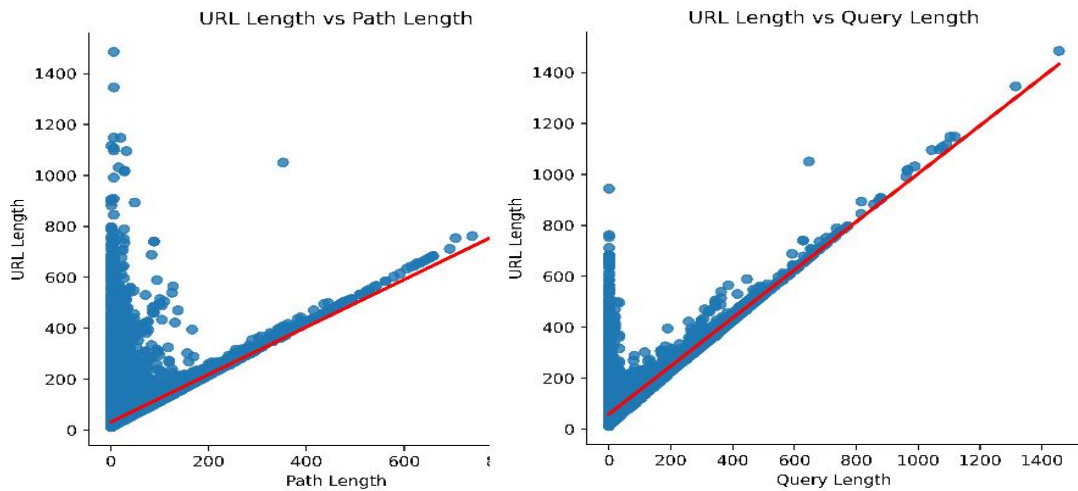


Figure 4.7: Path and Query vs URL Length Correlation plot

We randomly selected 1000 samples from each month and generated graphs to assess the correlations between different components and the overall URL length. Our findings revealed that the authority length had minimal impact, exhibiting a correlation coefficient of 0.2406. Conversely, both path and query lengths exhibited positively moderate relationships, with correlation coefficients of 0.645 and 0.699, respectively. This finding corresponds with the expectation that query plays a role in increasing

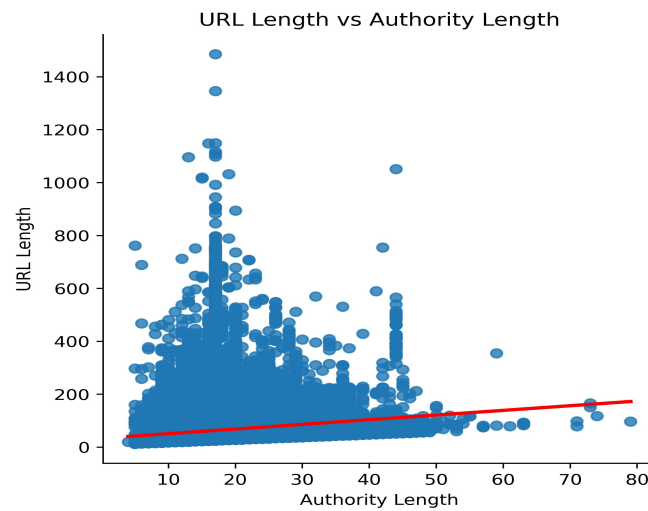


Figure 4.8: Authority vs URL Length Correlation plot

URL lengths. However, it was surprising to discover that the path component equally contributes significantly to the lengthening of URLs, which was not initially anticipated.

To encapsulate and reinforce our identified trends and patterns, we have constructed a stacked bar chart displaying the average changes in authority, path, and query lengths over time. The authority length remains relatively consistent and hasn't shown significant variation. On the other hand, there is a slight increase in both path and query lengths, indicating their potential contribution to the overall growth in URL lengths. Excessive values in the query and path components are primarily due to the presence of encoded characters, timestamps, and encrypted strings (session keys, other identifiers). This phenomenon is predominantly observed in URIs from the most recent years.

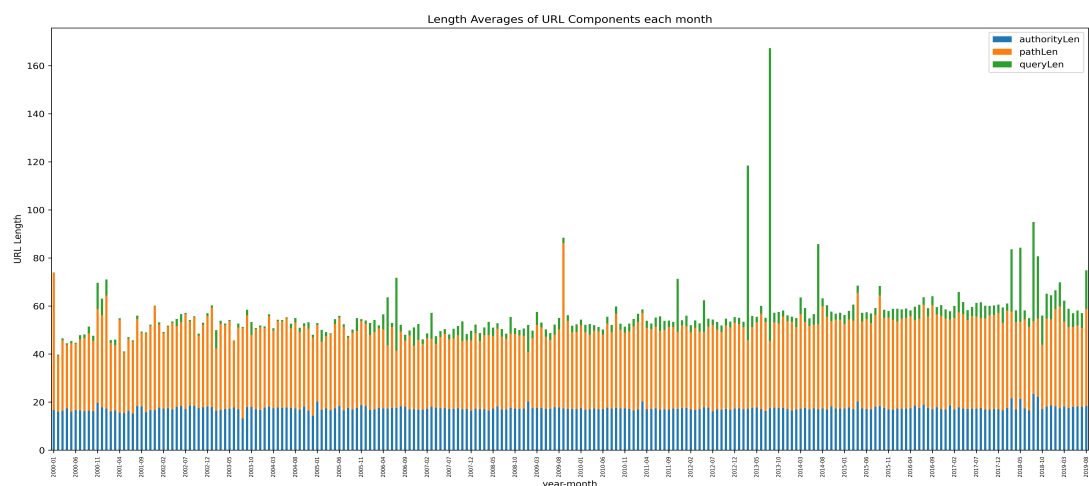


Figure 4.9: Stacked Bar Chart showing changes over Time

Chapter 5

Conclusions

In conclusion, our comprehensive analysis of URL characteristics and their constituents has provided valuable insights into the evolving nature of web addresses over time. We embarked on this journey by pre-processing and aggregating an extensive dataset encompassing years of internet history. Through meticulous data cleaning, transformation, and exploration, we unraveled patterns that shed light on how URLs have transformed.

Our findings revealed that the length of URLs has indeed shown an upward trend over the years, attributed to the inclusion of query and path components. This growth is notably evident in query length. Surprisingly, the path, initially considered a minor contributor, has also played a significant role in extending URL lengths. Our Time-series graphs, box-plots, histograms, and scatter plots have effectively highlighted these relationships and trends, providing a comprehensive insight into the evolution of URLs.

We uncovered outliers containing special characters, spaces, timestamps, and encryption string. We also found that newer URLs contain text in various languages, which serves as evidence for the increasing internationalization of the web. These outliers have notably skewed the overall data distribution in particular years. These anomalies emphasize the dynamic and evolving nature of web addresses, as modern internet practices have given rise to these unique patterns.

In summary, this study contributes to a deeper comprehension of URL characteristics and their evolution over the past decades. The trends and patterns we have identified hold relevance not only for historical analysis but also for contemporary web practices. By delving into the complex interplay of URL components and their impact on length, we have provided valuable insights into how internet practices have transformed and shaped the landscape of web addresses.

5.1 Future Works

Numerous avenues for future research are available. Given the dominance of August 2019 data, a comparative analysis with archives from various years could provide further insights into trends. Additionally, utilizing indices instead of numerous WARC files from different years might offer efficiency gains due to their reduced size. Exploring the depth of web pages and the evolution of query element counts over time could provide valuable context. Furthermore, delving into the specific special characters and strings that are shaping modern addresses would offer an intriguing perspective on contemporary internet practices.

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